

Observables and efficiencies for the
21-09-2026-full-vs-report-v1 study
A first coherent account of what the controller does to the population

Offline analysis, MA-CC

2026-09-23

Contents

1	What this report is	2
2	Design, data and the unit of independence	2
2.1	Design	2
2.2	The independent unit	2
3	Effects on the vote: shared and directional components	3
3.1	Definitions	3
3.2	What the data show	3
3.3	How the effects build with the horizon	5
4	One-round response: observational and causal	6
5	The command channel: does the population hear which answer was requested?	7
6	Endpoint and trajectory information	9
6.1	Estimators	9
6.2	Trajectory cost and its decomposition	10
6.3	Why η_{task} is mostly unavailable	12
7	Null diagnostics	12
8	Evidence: what the controller does to what agents know	12
9	Does the controller persuade agent by agent?	14
9.1	Two features of the board that shape the test	14
9.2	The estimator	15
9.3	Result	16
9.4	What this does and does not establish	16
10	The account	16
11	What would falsify this, and what is not established	17

1 What this report is

This is an offline analysis of the archived study `21-09-2026-full-vs-report-v1`. No simulation was run and no provider request was made; the source directory was opened read-only. Every quantity follows the definitions in `shared_references/BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES.md`, and the section numbers quoted below are that document’s.

The purpose is narrative, not just tabulation: to say what the controller actually does to this population, and which of the many available numbers are load-bearing for that story. Section 10 states the account; everything before it is the evidence, and Section 11 is what would falsify it.

2 Design, data and the unit of independence

2.1 Design

A population of $N = 24$ LLM agents discusses one fixed MuSR team-allocation task (`task_003`) on a shared message board for 15 rounds. Truth is `ALLOCATION_0`; the designated false target is `ALLOCATION_2`; `ALLOCATION_1` is the remaining wrong answer. Writing $n_{a,t}$ for the number of agents voting for allocation a after round t ,

$$Y_t = (n_{0,t}, n_{1,t}, n_{2,t}), \quad x_{a,t} = n_{a,t}/N, \quad \Gamma_h = (Y_0, Y_1, \dots, Y_h), \quad (1)$$

with Y_0 the initialization state. Counts sum to N at every round; this was checked on every trajectory rather than assumed.

Two experimental factors distinguish this study from the earlier paired-parent archive, and both matter for the interpretation:

Communication profile. *Report-only* (RO) permits only REPORT messages; *full communication* (FC) permits the richer message set. This changes what the *population* can do among itself, not only what the controller can do.

Randomized activation. The controller’s decision to act in round t is drawn with a logged probability $e_t = P(U_t = 1 \mid \text{pre-action information})$, a sigmoid with threshold 0.5 and slope 4. Realized e_t lies in $[0.119, 0.881]$, bounded away from 0 and 1.

That second point is the important methodological gain over the paired archive. Because the gate is genuinely randomized with known and non-degenerate propensities, the inverse-probability-weighted causal estimator of reference section 3 is *available* here. It was not available for a deterministic always-policy. We therefore report a causal one-round response alongside the observational one, and they can be compared.

The first physical round carries no controller decision by design; rounds 1..14 do. Posts satisfy $c_t = bU_t$ exactly in every trajectory, with $b \in \{3, 12, 18\}$, which was verified rather than assumed.

2.2 The independent unit

The archive holds 1382 complete trajectories, all of which passed validation (0 exclusions). They come from only 60 distinct physical initializations, reused across settings and arms. **The initialization is the only independent unit.** Neither the 1382 trajectories, nor the 24 votes inside a population, nor the rounds within a trajectory are independent replicates. Every split and every bootstrap in this report resamples whole initializations with all their arms attached.

A *comparison* fixes a communication profile, a persistence ρ and a budget b , and retains the initializations having all three arms — silence, the truth request and the false request — complete. This yields 9 comparisons with m between 49 and 60 (Table 1). The four full-communication/ $\rho=1$ cells with a false target are absent from the archive, so no two-target comparison exists there; that block is simply missing, and is not imputed.

Table 1: Initializations with a complete silent / target-0 / target-2 triple.

communication profile	ρ	b	initializations m
full communication	0.75	3	60
full communication	0.75	12	53
full communication	0.75	18	49
report only	0.75	3	59
report only	0.75	12	51
report only	0.75	18	53
report only	1.00	3	56
report only	1.00	12	55
report only	1.00	18	52

The silent arm carries no budget, so one silent trajectory is shared by the three budgets within a setting. That shared randomness is real and is preserved by resampling whole initializations.

3 Effects on the vote: shared and directional components

3.1 Definitions

For a measured answer a , the paired policy susceptibility of reference section 3 is the finite contrast between running the whole procedure and running nothing,

$$G_h^{\pm,a} = \mathbb{E}[x_{a,h}^{\pm} - x_{a,h}^{\text{base}}], \quad (2)$$

where $+$ requests truth and $-$ requests the false target. This is not a derivative with respect to budget. Reference section 4 splits it into a shared component and a directional half-contrast,

$$M_h^{(a)} = \frac{1}{2}(G_h^{+,a} + G_h^{-,a}), \quad D_h^{(a)} = \frac{1}{2}(G_h^{+,a} - G_h^{-,a}), \quad (3)$$

so $G_h^{\pm,a} = M_h^{(a)} \pm D_h^{(a)}$ and the full effect of switching the requested target is $2D_h^{(a)}$. We export both conventions. M answers “what does having a controller at all do?”; D answers “does it matter which answer we asked for?”. The identities $\sum_a M_h^{(a)} = \sum_a D_h^{(a)} = 0$ and $M^{(\text{err})} = -M^{(0)}$ were verified numerically.

3.2 What the data show

Figure 1 is the single most informative picture in this report, and it already contains the main result. Under silence the population is *good at this task*: the report-only baseline reaches 95.0% truth by $h = 15$ and the full-communication baseline 68.7%. Control almost always moves it away from that.

Three facts stand out in Table 2 and Figure 2.

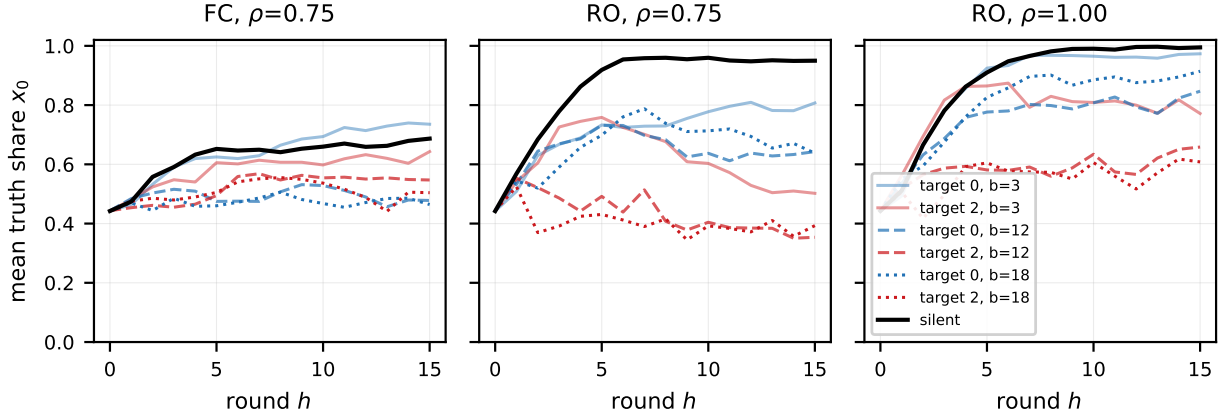


Figure 1: Mean truth share against round, by setting. Black: silence. Blue: truth requested. Red: false target requested. Line style encodes the posting budget. The silent curve is the same in every panel of a setting because the silent arm is shared across budgets.

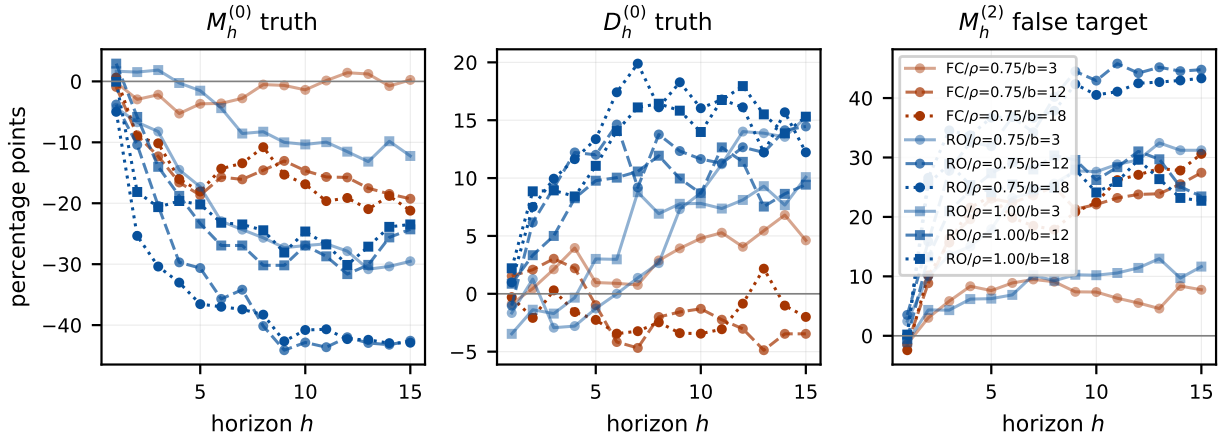


Figure 2: Shared component M and directional half-contrast D against horizon, in percentage points. Left and centre: truth coordinate. Right: the false-target coordinate.

Table 2: Effects on the vote at $h = 15$, in percentage points. “base” is the mean silent truth share. $D^{(0)}$ carries its 95% paired bootstrap interval.

prof.	ρ	b	m	base	$G^{+,0}$	$G^{-,0}$	$M^{(0)}$	$D^{(0)}$ [95%]	$M^{(2)}$	$D^{(2)}$
FC	0.75	3	60	68.7	4.9	-4.4	0.2	4.6 [-1.3, 10.4]	7.7	-5.5
FC	0.75	12	53	70.5	-22.7	-15.8	-19.3	-3.5 [-11.0, 4.6]	27.4	2.5
FC	0.75	18	49	69.6	-23.2	-19.2	-21.2	-2.0 [-9.3, 5.0]	30.6	1.7
RO	0.75	3	59	95.0	-14.3	-44.8	-29.5	15.3 [9.5, 21.3]	31.2	-14.9
RO	0.75	12	51	92.4	-28.1	-57.0	-42.6	14.5 [8.4, 20.4]	44.8	-14.4
RO	0.75	18	53	94.4	-30.7	-55.1	-42.9	12.2 [7.1, 17.4]	43.3	-11.9
RO	1.00	3	56	99.5	-2.2	-22.3	-12.2	10.1 [6.7, 13.5]	11.6	-9.2
RO	1.00	12	55	99.5	-14.8	-33.7	-24.3	9.4 [5.4, 13.3]	23.4	-8.9
RO	1.00	18	52	99.6	-8.2	-38.8	-23.5	15.3 [11.5, 19.3]	22.7	-14.8

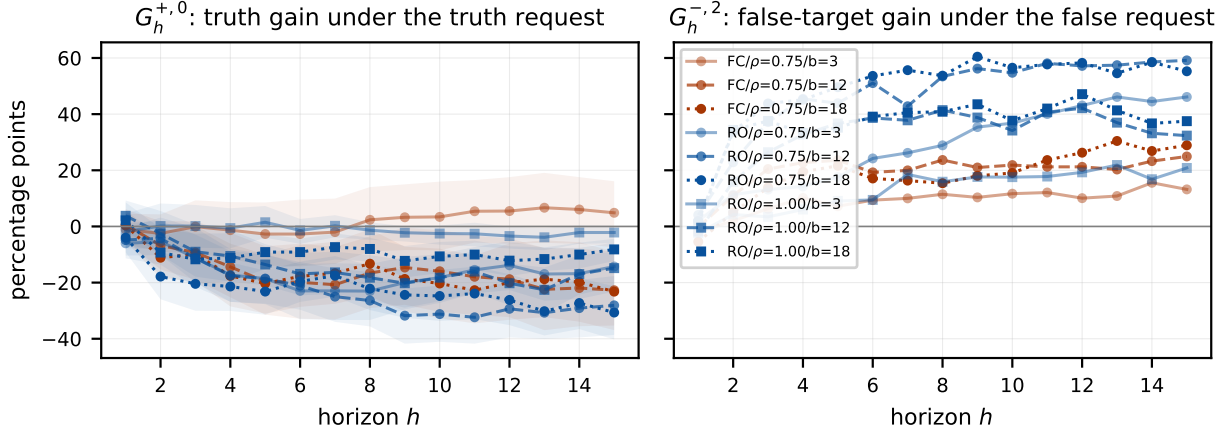


Figure 3: Own-target gains against horizon with 95% paired-initialization bootstrap bands: truth gain when truth is requested (left) and false-target gain when the false target is requested (right).

The shared component is negative almost everywhere, and grows with budget. $M_{15}^{(0)}$ is negative in 8 of the 9 comparisons, ranging from -42.9 to 0.2 points. Within the report-only setting at $\rho = 0.75$ it deepens monotonically with the budget: -29.5 points at $b = 3$ and -42.9 at $b = 18$. Posting more does not steer better; it damages more. The corresponding $M^{(2)}$ is positive throughout, so the mass leaving truth arrives largely at the false target regardless of which target was requested.

But the directional component is real, and larger than in the paired archive. $D_{15}^{(0)}$ is positive in 7 of 9 comparisons, spanning -3.5 to 15.3 points. In report-only at $b = 3$, $D^{(0)} = 15.3$ points with a bootstrap interval excluding zero. Requesting truth really does preserve more truth than requesting the false target, even though both requests are harmful on net. The controller is not merely a target-independent disturbance here — which is what the paired always-policies looked like — but a disturbance with a genuine directional component riding on top.

One cell is different, and it is the informative one. Full communication at $\rho = 0.75$, $b = 3$ has $G_{15}^{+,0} = +4.9$ points (95% interval $[-7.4, 16.1]$) and $M^{(0)} \approx 0$: at low budget, in the setting where the population is *worse* on its own (68.7% versus 95.0%), a truth-targeted controller is roughly neutral to mildly helpful. Raise the budget in that same setting and the benefit disappears: $M^{(0)} = -21.2$ points at $b = 18$. Helpfulness is confined to the corner where the population has room to improve and the controller speaks quietly.

3.3 How the effects build with the horizon

Because every quantity in this report is computed at each h , the horizon axis is itself a finding rather than a presentational choice. Three patterns recur across Figures 1, 2, 3 and 5.

Nothing happens immediately. At $h = 1$ both $M^{(0)}$ and $D^{(0)}$ are within a point or two of zero in every comparison. One round of posting does not move a population of 24 agents measurably. Any analysis that stopped at the first horizon would conclude the controller does nothing.

Damage accumulates without saturating. $M^{(0)}$ falls steadily across the whole range and is still falling at $h = 15$ in the larger-budget comparisons. This is the shape of a process that erodes rather than one that shifts an equilibrium. Figure 1 makes the mechanism visible in the report-only panels: the controlled branches track the silent curve upward for roughly the first five rounds, reach a peak, and only then decline. The controller does not prevent the population from finding the answer; it degrades a consensus the population has already reached.

Direction saturates early. $D^{(0)}$ rises steeply to about $h = 7$ and then flattens or drifts. Command information $I(Z; V_h)$ behaves the same way. So the target-specific part of the signal is delivered in the first half of the run and does not grow after that, while the target-independent damage keeps compounding. That divergence in timescales is the cleanest dynamical statement this archive supports, and it is what drives the efficiency ratios down at long horizons: the numerator plateaus while the denominator grows.

A practical consequence: the measured “efficiency” of this controller is a function of when you stop looking. Quoting any of these ratios without its horizon is meaningless.

4 One-round response: observational and causal

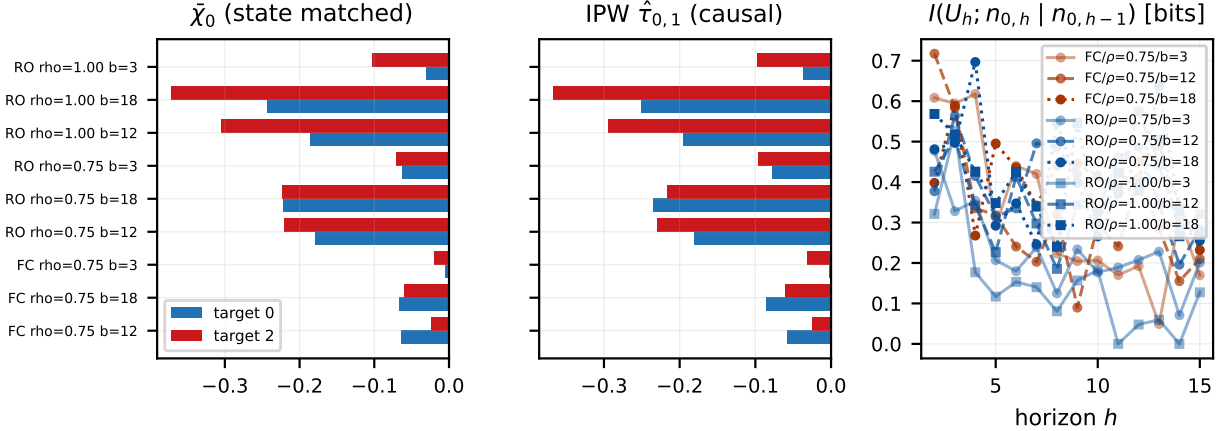


Figure 4: Left: state-matched response $\bar{\chi}_0$, weighted by visited states. Centre: the inverse-probability-weighted causal response $\hat{\tau}_{0,1}$. Right: one-round activation information against horizon.

Because activation is randomized with logged propensities, we can ask the causal question directly: what does forcing the controller to act *this* round do to truth support? The estimator is

$$W_t = \frac{U_t}{e_t} - \frac{1 - U_t}{1 - e_t}, \quad \hat{\tau}_{a,\ell} = \frac{1}{L_\ell} \sum_t W_t (x_{a,t+\ell} - x_{a,t}). \quad (4)$$

Realized activation rates are 0.20 to 0.72, and the largest weight is 8.4, so no single round dominates. At lag 1 the causal truth response is negative in 9 of the 9 truth-request rows, and the range across all rows is -0.367 to -0.002. By lag 5 it is -0.132 to +0.047: the damage accumulates rather than reverting.

This is worth pausing on. The observational state-matched contrast $\bar{\chi}_0$ and the causal $\hat{\tau}_{0,1}$ agree in sign and rough magnitude. In the paired archive we could only compute the observational version and had to caveat it heavily. Here the randomized gate lets us say the stronger thing: *acting*

Table 3: Response families for the truth coordinate. $P(U=1)$ is the realized activation rate; $\bar{\chi}_0$ is the state-matched contrast; $\hat{\tau}_{0,\ell}$ is the IPW causal response at lag ℓ ; “avail.” is the cell-ratio available-susceptibility; $\max |W|$ is the largest inverse-probability weight.

prof.	ρ	b	z	$P(U=1)$	$\bar{\chi}_0$	uncond.	$\hat{\tau}_{0,1}$	$\hat{\tau}_{0,5}$	avail.	$\max W $
FC	0.75	3	0	0.380	-0.0054	0.0283	-0.0018	-0.0658	0.0015	8.4
FC	0.75	3	2	0.605	-0.0195	-0.0346	-0.0317	-0.0089	-0.0750	8.4
FC	0.75	12	0	0.446	-0.0635	-0.0026	-0.0574	-0.0356	-0.0763	8.4
FC	0.75	12	2	0.487	-0.0229	-0.0481	-0.0248	0.0215	-0.0536	8.4
FC	0.75	18	0	0.420	-0.0658	-0.0385	-0.0848	-0.0307	-0.1492	8.4
FC	0.75	18	2	0.458	-0.0592	-0.0841	-0.0597	0.0030	-0.1018	8.4
RO	0.75	3	0	0.317	-0.0625	-0.0135	-0.0771	-0.1221	-0.2248	8.4
RO	0.75	3	2	0.608	-0.0697	-0.1128	-0.0962	-0.1322	-0.2358	8.4
RO	0.75	12	0	0.329	-0.1784	-0.0982	-0.1801	-0.1052	-0.3913	8.4
RO	0.75	12	2	0.434	-0.2205	-0.2841	-0.2291	-0.0365	-0.4079	8.4
RO	0.75	18	0	0.314	-0.2209	-0.1286	-0.2352	-0.0927	-0.5550	8.4
RO	0.75	18	2	0.418	-0.2233	-0.3057	-0.2162	-0.0093	-0.3564	8.4
RO	1.00	3	0	0.195	-0.0298	0.0214	-0.0359	-0.0365	-0.1382	8.4
RO	1.00	3	2	0.718	-0.1020	-0.1635	-0.0975	-0.0060	-0.4032	8.4
RO	1.00	12	0	0.269	-0.1858	-0.0976	-0.1951	-0.0467	-0.5405	8.4
RO	1.00	12	2	0.556	-0.3050	-0.3842	-0.2941	0.0467	-0.6991	8.4
RO	1.00	18	0	0.228	-0.2423	-0.1601	-0.2504	-0.0167	-0.7769	8.4
RO	1.00	18	2	0.530	-0.3720	-0.4757	-0.3672	0.0366	-0.8213	8.4

causally reduces truth support on the very next round, and this is not an artifact of the controller choosing to act when the population was already drifting.

The right panel of Figure 4 shows the one-round activation information $I(U_h; n_{0,h} \mid n_{0,h-1})$ of reference section 6, which reads between 0.127 and 0.321 bits at $h = 15$. **These numbers should not be interpreted as transmitted information.** They are frequency plug-in estimates of a conditional mutual information computed from $m \approx 60$ observations spread over many $(n_{0,h-1}, n_{0,h})$ cells, so most cells hold one or two observations and the estimator is dominated by upward bias — the same pathology Table 5 exhibits for the endpoint, where the in-sample plug-in overstates the held-out score by 0.252 bits. The declining trend with horizon is also consistent with bias rather than signal: as the population converges, the conditioning variable takes fewer distinct values and the bias falls. We report the quantity because the reference defines it, and we decline to read anything into its magnitude. The honest one-round statements in this archive are the state-matched and IPW responses in the left two panels, which are differences of means rather than nonlinear functionals and do not carry this bias.

5 The command channel: does the population hear which answer was requested?

Reference section 9 defines the single-agent command channel by drawing $Z \in \{0, 2\}$ balanced, running the target-conditioned policy, and reading one uniformly chosen agent’s final vote V_h :

$$p_h(v \mid z) = \mathbb{E}[x_{v,h}^z], \quad I(Z; V_h) = \frac{1}{2} \sum_z \sum_v p_h(v \mid z) \log_2 \frac{p_h(v \mid z)}{\bar{p}_h(v)}, \quad (5)$$

with $\eta_{\text{command}} = I(Z; V_h)/\mathcal{H}(Z)$ numerically equal to the information in bits because $\mathcal{H}(Z) = 1$. Following is $F_h = \frac{1}{2}[p_h(0|0) + p_h(2|2)]$ against the matched silent benchmark $F_h^{\text{base}} = \frac{1}{2}\mathbb{E}[x_{0,h}^{\text{base}} + x_{2,h}^{\text{base}}]$.

Marginalizing a uniformly chosen agent is exact from the vote fractions, so no classifier is used here and none is needed.

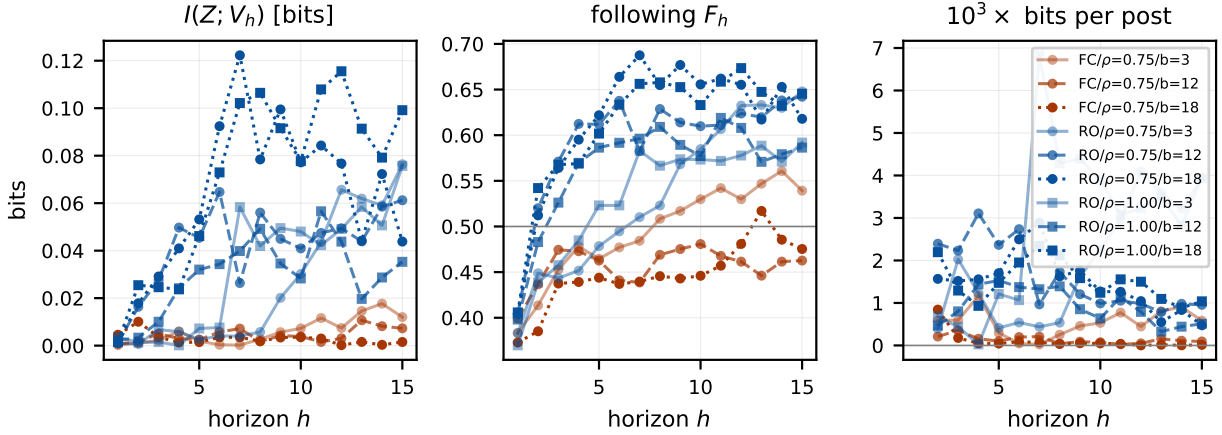


Figure 5: Command information, following probability and information per post against horizon.

Table 4: Command transmission and resources at $h = 15$. Posts and senses are averaged over the two target branches. $\mathcal{E}_\lambda$ is bits per post-equivalent with sensing charged at λ .

prof.	ρ	b	$I(Z; V_h)$	F_h	$F_h - F_h^{\text{base}}$	posts	senses	$10^3 \mathcal{E}_{\lambda=0}$	$10^3 \mathcal{E}_{\lambda=1}$
FC	0.75	3	0.0120	0.539	0.090	20.7	149	0.581	0.0710
FC	0.75	12	0.0073	0.463	0.011	78.3	150	0.093	0.0320
FC	0.75	18	0.0015	0.475	0.028	110.6	147	0.014	0.0060
RO	0.75	3	0.0764	0.648	0.159	19.4	168	3.933	0.4078
RO	0.75	12	0.0612	0.642	0.155	64.1	168	0.955	0.2639
RO	0.75	18	0.0438	0.618	0.123	92.2	168	0.475	0.1685
RO	1.00	3	0.0757	0.592	0.093	19.2	168	3.946	0.4045
RO	1.00	12	0.0353	0.586	0.087	69.3	168	0.510	0.1488
RO	1.00	18	0.0992	0.645	0.147	95.5	168	1.038	0.3763

Command information at $h = 15$ ranges from 0.0015 to 0.0992 bits, and following from 0.463 to 0.648. In report-only at $b = 3$, $I(Z; V_{15}) = 0.0764$ bits with $F = 0.648$ against a silent benchmark, an improvement of +0.159.

The exact identity of reference section 9,

$$F_h - F_h^{\text{base}} = \frac{1}{2}(G_h^{+,0} + G_h^{-,2}), \quad (6)$$

explains why following can improve while truth collapses: it is the average of the two *own-target* gains, and the false branch’s large gain on its own target carries it. A controller that is very good at installing the false answer, and mildly bad at defending the true one, still scores well on “following”. Following is not a measure of benefit.

Sensing is charged honestly. The controller reads sampled votes on every gated round whether or not it posts, so sensing expenditure is the same in all controlled branches and is not a cost specific to a sensing policy. The $\lambda = 1$ column of Table 4 shows what happens to apparent efficiency once those reads are priced at one post each: the numbers drop by roughly an order of magnitude without anything having changed about what was transmitted.

6 Endpoint and trajectory information

6.1 Estimators

$I(Z; Y_h)$ has 325 possible endpoint states and cannot be read off a frequency table at $m \approx 50$. Following reference section 12 we fit a balanced target classifier $g(Z | Y_h)$ and score the held-out

$$\hat{L}_I(h) = \mathbb{E}[\log_2 2g(Z | Y_h)], \quad (7)$$

whose population gap from $I(Z; Y_h)$ is the expected KL between the true and fitted posteriors — a lower bound in expectation, with negative realized values permitted. Candidates are a constant prior, L2 logistic regression on the two vote fractions, quadratic logistic features, and a one-hidden-layer MLP of width 4 or 8; selection is by nested held-out log loss inside training initializations only.

Table 5: Frequency plug-in versus held-out estimates of $I(Z; Y_{15})$, in bits.

prof.	ρ	b	freq. $\alpha=0$	freq. $\alpha=1$	held out $\alpha=1$	held out $\alpha=10$	classifier	selected
FC	0.75	3	0.180	0.174	-0.527	-0.283	0.005	mlp-w4-l0.1
FC	0.75	12	0.238	0.230	-0.602	-0.307	-0.005	constant
FC	0.75	18	0.312	0.301	-0.404	-0.176	-0.002	constant
RO	0.75	3	0.374	0.364	-0.270	-0.043	0.087	linear-l0.1
RO	0.75	12	0.362	0.352	-0.420	-0.135	0.108	quadratic-l0.1
RO	0.75	18	0.379	0.368	-0.483	-0.167	0.075	linear-l0.1
RO	1.00	3	0.392	0.381	0.214	0.231	0.145	mlp-w4-l0.01
RO	1.00	12	0.347	0.337	-0.508	-0.202	0.119	quadratic-l0.1
RO	1.00	18	0.443	0.430	-0.095	0.054	0.222	quadratic-l0.1

Table 5 is a warning about the frequency estimator. The in-sample plug-in averages 0.336 bits; the held-out classifier averages 0.084, a gap of 0.252 bits of pure optimism. With nearly unique endpoints the raw frequency MI is mostly counting its own noise. Only the held-out column should be quoted.

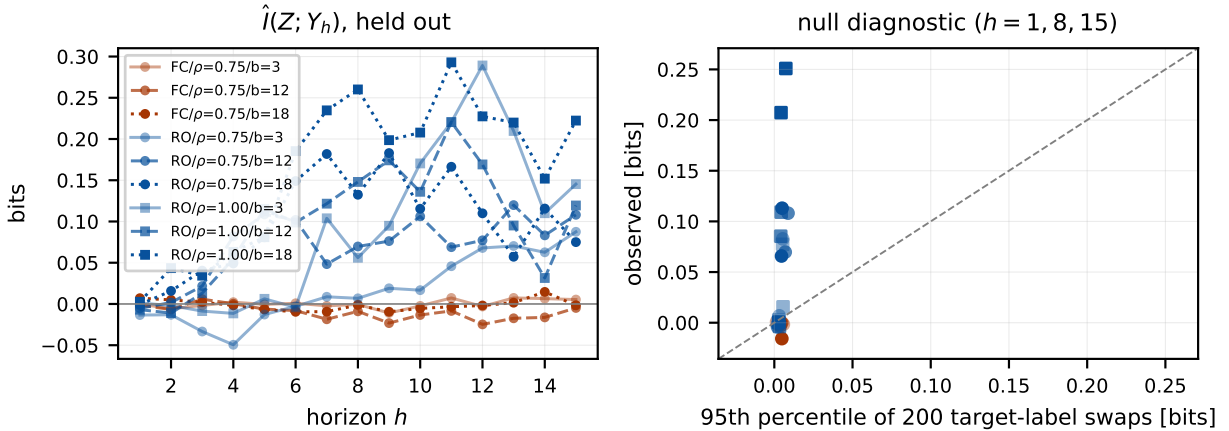


Figure 6: Left: held-out endpoint target information against horizon. Right: observed score against the 95th percentile of 200 within-initialization target-label swaps.

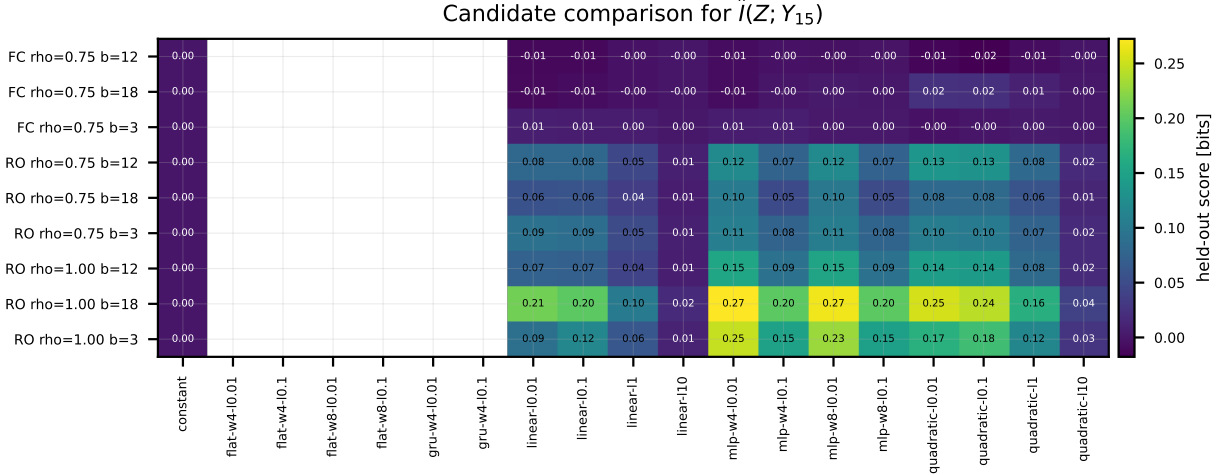


Figure 7: Held-out score of every candidate for $\hat{I}(Z; Y_{15})$.

6.2 Trajectory cost and its decomposition

For the divergence terms we fit balanced controlled-versus-silent critics and report the NWJ objective $\mathbb{E}_P f - \mathbb{E}_{\text{base}} e^f + 1$ and the Donsker–Varadhan objective $\mathbb{E}_P f - \log \mathbb{E}_{\text{base}} e^f$, both population lower bounds in nats, with a capped critic and baseline-tail diagnostics. Candidates add path summaries, a flat MLP over the whole path, and small GRUs. The target-averaged cost and its exact decomposition (reference section 11) are

$$\mathcal{C}_\pi = \frac{1}{2}K_\pi^0 + \frac{1}{2}K_\pi^2 = I_{\text{nat}}(Z; \Gamma) + D_{\text{KL}}(Q \| P_{\text{base}}), \quad Q = \frac{1}{2}P_\pi^0 + \frac{1}{2}P_\pi^2, \quad (8)$$

with the data-processing chain

$$I_{\text{nat}}(Z; V_h) \leq I_{\text{nat}}(Z; Y_h) \leq I_{\text{nat}}(Z; \Gamma) \leq \mathcal{C}_\pi \quad (9)$$

and the efficiencies $\eta_{\text{end}} = I_{\text{nat}}(Z; Y_h)/\mathcal{C}_\pi$, $\eta_{\text{ctl}} = I_{\text{nat}}(Z; \Gamma)/\mathcal{C}_\pi$. The three terms are fitted independently, so their sum need not reconcile exactly; the gap is a diagnostic.

Table 6: Divergences, decomposition and efficiencies at $h = 15$, in nats. Entries are blank where the cost bootstrap is not bounded away from zero, or, for η_{task} , where K_{min} is not identified from the silent sample.

prof.	ρ	b	K^0	K^2	$\mathcal{C}_\pi$	$I(Z; \Gamma)$	D_Q	sum	$I(Z; Y)$	η_{end}	η_{ctl}	η_{task}
FC	0.75	3	-0.003	0.024	0.010	0.003	-0.020	-0.017	0.004	—	—	—
FC	0.75	12	0.205	0.176	0.190	-0.015	0.174	0.159	-0.003	—	—	—
FC	0.75	18	-0.153	-0.921	-0.537	-0.005	-0.420	-0.425	-0.001	—	—	—
RO	0.75	3	0.465	1.791	1.128	0.046	0.567	0.613	0.061	—	—	—
RO	0.75	12	0.770	2.729	1.750	0.167	1.477	1.644	0.075	0.043	0.095	0.675
RO	0.75	18	1.409	2.182	1.795	0.123	1.761	1.884	0.052	—	—	—
RO	1.00	3	0.080	3.506	1.793	0.271	1.363	1.633	0.101	0.056	0.151	—
RO	1.00	12	2.119	4.694	3.406	0.157	2.970	3.127	0.083	0.024	0.046	—
RO	1.00	18	2.293	4.761	3.527	0.276	2.939	3.215	0.154	0.044	0.078	—

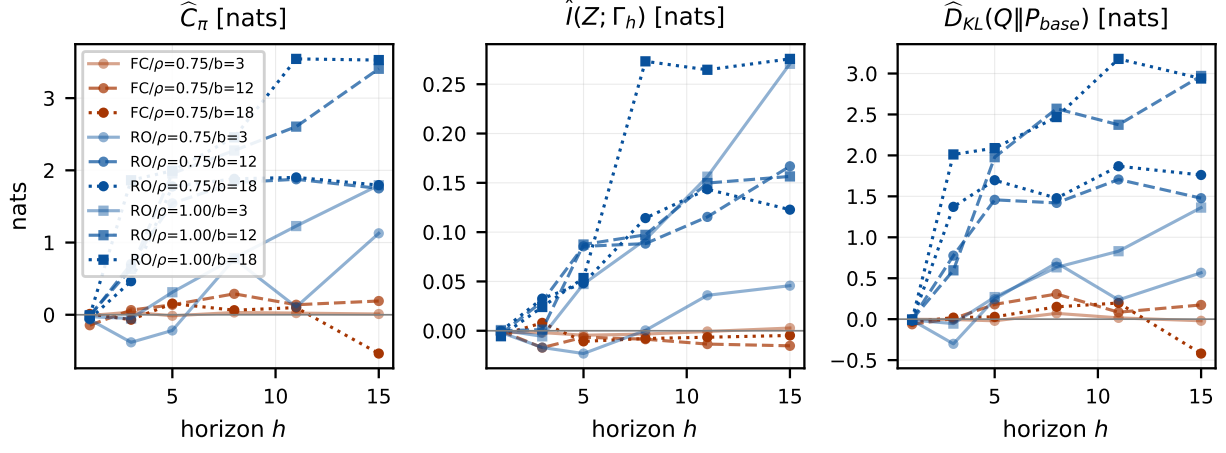


Figure 8: Trajectory cost and the two terms of its decomposition against horizon.

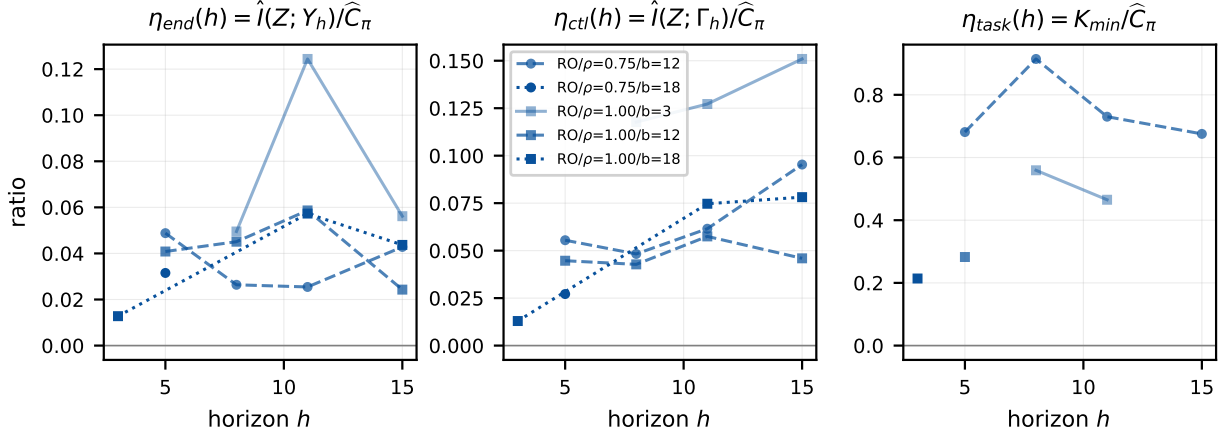


Figure 9: Efficiency ratios against horizon, shown only where no bootstrap draw of the cost is non-positive.

6.3 Why η_{task} is mostly unavailable

Reference section 11 defines the minimum displacement needed to move a reward by δ as the Legendre transform of the baseline log moment generating function, $K_{\min}(\delta) = \sup_{\lambda} [\lambda\delta - \psi(\lambda)]$, and $\eta_{\text{task}} = K_{\min}(\delta)/K_{\pi} \leq 1$.

Computed naively this returns numbers like 33, 66 and 158 nats for the report-only $\rho = 1$ comparisons, which would give $\eta_{\text{task}} \approx 10$ — impossible for a quantity bounded by one. The cause is a support failure, not a large cost. Across all initializations of that setting the silent truth share never falls below 0.917; the controlled arms shift the mean to below that. The requested mean therefore lies outside the observed baseline range, the empirical moment generating function has no mass there, and the optimal tilt runs to whatever bound the λ grid happens to impose. The reported value is then an artifact of the grid, not an estimate.

We detect this explicitly — requested mean inside the observed baseline support, and the optimizer interior rather than at the grid edge — and report $K_{\min}$ as unidentified when either test fails. It is identified in 46 of 54 rows, and combined with the requirement that the cost bootstrap be bounded away from zero this leaves 8 usable η_{task} values. This is the same missing-baseline-event problem the reference warns about, and it is worth stating plainly: *the controller drives this population somewhere silence never goes*, and that is precisely why the information-theoretic minimum cost cannot be estimated from the silent sample.

7 Null diagnostics

Swapping the two target branches within an initialization destroys target identity while preserving everything else about the pair. Under that exchangeability null the observed endpoint score should sit inside the permutation distribution. 12 of 27 cells exceed their null at the 5% level, and at $h = 15$ it is 6 of 9. This is an exchangeability diagnostic, not a proof of causality: it says the target label carries information the vote endpoint can see.

8 Evidence: what the controller does to what agents know

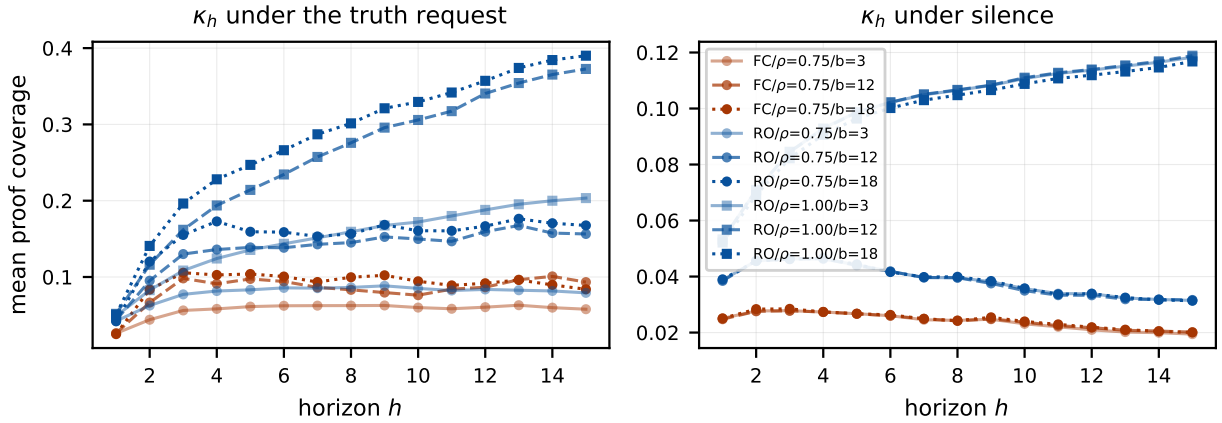


Figure 10: Mean active supporting-fact coverage κ_h against horizon, under the truth request and under silence.

Table 7: Within-initialization target-label swaps (200 per cell, full refit).

comparison	h	observed	null mean	null p95	p
full communication rho0.75 b12	1	-0.001	-0.001	0.003	0.687
full communication rho0.75 b12	8	-0.003	-0.002	0.004	0.806
full communication rho0.75 b12	15	0.000	-0.002	0.005	0.537
full communication rho0.75 b18	1	-0.016	-0.002	0.005	0.955
full communication rho0.75 b18	8	0.000	-0.002	0.004	0.478
full communication rho0.75 b18	15	-0.002	-0.002	0.003	0.756
full communication rho0.75 b3	1	0.001	-0.002	0.001	0.080
full communication rho0.75 b3	8	-0.001	-0.001	0.004	0.617
full communication rho0.75 b3	15	-0.002	-0.001	0.006	0.701
report only rho0.75 b12	1	-0.004	-0.001	0.002	0.900
report only rho0.75 b12	8	0.070	-0.002	0.007	0.005
report only rho0.75 b12	15	0.108	-0.001	0.009	0.005
report only rho0.75 b18	1	0.000	-0.001	0.003	0.194
report only rho0.75 b18	8	0.113	-0.001	0.005	0.005
report only rho0.75 b18	15	0.066	-0.002	0.005	0.005
report only rho0.75 b3	1	-0.001	-0.001	0.002	0.716
report only rho0.75 b3	8	0.007	-0.001	0.003	0.015
report only rho0.75 b3	15	0.083	-0.001	0.005	0.005
report only rho1.00 b12	1	-0.004	-0.000	0.003	0.910
report only rho1.00 b12	8	0.085	-0.001	0.004	0.005
report only rho1.00 b12	15	0.109	-0.002	0.004	0.005
report only rho1.00 b18	1	0.001	-0.001	0.003	0.134
report only rho1.00 b18	8	0.251	-0.003	0.007	0.005
report only rho1.00 b18	15	0.207	-0.003	0.004	0.005
report only rho1.00 b3	1	0.003	-0.001	0.003	0.055
report only rho1.00 b3	8	0.016	-0.003	0.006	0.020
report only rho1.00 b3	15	0.077	-0.001	0.005	0.005

Reference section 5 defines κ_t as mean active proof coverage and ϕ_t as the share of agents holding a full configured proof. This section contains the result that most constrains the mechanism, and it is not the one we expected.

The controller substantially increases evidence coverage. At $h = 15$, mean κ is 0.056 under silence, 0.178 under the truth request and 0.238 under the false request. Per comparison the truth-request arm carries 1.72 to 5.35 times the silent coverage, rising with the posting budget. The controller is not starving the population of information. It is demonstrably *delivering* verified facts, and delivering more of them as it posts more.

The false-target controller delivers about as much as the truth-target one. $\kappa_{t2} \geq \kappa_{t0}$ at the final horizon in 8 of the 9 comparisons, and the exception is close to a tie. This is exactly what the design implies and is easy to miss: the controller posts *true* facts drawn from the task pool together with a vote for its assigned target. A true fact accompanied by a vote for `ALLOCATION_2` still raises measured proof coverage. Coverage counts facts held, not conclusions correctly drawn.

No agent ever completes a proof. ϕ_h is exactly zero (maximum observed 0.0 over all 135 arm-horizon rows) in every arm, comparison and horizon, including under silence. Not one of the 24 agents ever holds a full configured proof at any point in this archive. The population is voting on fragments throughout.

Those three facts together rule out the simplest explanation of the damage — that the controller crowds out peer information — and point at a different one. Evidence supply goes *up* while truth support goes *down*. What the controller adds alongside the facts is a salient, repeated, explicit vote; and because no agent can verify a conclusion from a complete proof, that vote is cheap to copy relative to the cost of assembling fragments. The controller’s influence appears to run through the *vote* it attaches, not through the *evidence* it supplies, and the evidence it supplies is not sufficient to protect an agent against the vote.

That was our reading when only these observables were available. Section 9 carries out the message-level test it calls for — linking which agents read which posts to how they then voted — and the result does *not* support the simplest version of it. At the margin, within a population and a round, agents who read more of the controller were no likelier to adopt its target. The vote-copying story survives only in a collective form, in which the controller moves a few agents and the population carries the rest; it is not supported as direct per-message persuasion. The observables here constrain the mechanism; they do not identify it, and neither does the micro test.

9 Does the controller persuade agent by agent?

The evidence observables above constrain the mechanism but do not identify it. The micro-update records can go further: for each of the 497,520 agent updates they record that agent’s vote before and after, and the identifiers of the messages it actually read. This section reports that test. It returns a null, and the way it returns a null is informative.

9.1 Two features of the board that shape the test

First, **the board is erased every round**: `message_lifetime_rounds` is 1 in all cells and the maximum `surviving_message_count` over the whole archive is 0.0. About 21 messages are created

per round and the same number expire. Message dose therefore measures the current round’s output only; nothing accumulates.

Second, **agents update sequentially and post as they go**. The controller posts first, so its share of the board decays across the 24 update slots within a round (Figure 11, left). An agent updating in the first slots reads about 8.0 controller messages out of 8.9; one updating in the last slots reads 4.2 out of 12.0, because peers have filled the board meanwhile. The number of controller messages available is constant at about 10.2 throughout — what changes is competition for the agent’s fixed reading allowance.

That matters because the outcome moves with update order too. Both dose and the probability of ending the update on the controller’s target decline across slots, so any unadjusted dose-response will report an effect of ordering as though it were an effect of messages.

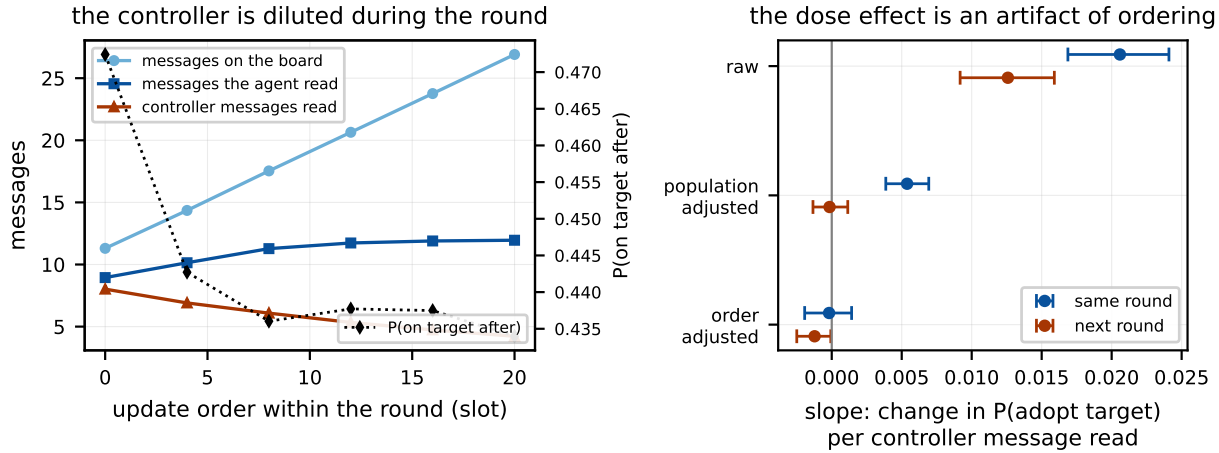


Figure 11: Left: within a round the controller’s messages are progressively crowded out of the agent’s fixed reading sample, and the outcome declines with them. Right: the estimated slope of adoption on controller messages read, before adjustment, after holding the population and round fixed, and after additionally removing the update-order trend. Bars are 95% cluster bootstrap intervals over 60 initializations.

9.2 The estimator

Restrict to agent updates where a controller message was on the board and the agent was *not* already voting the target, so the outcome is adoption rather than retention. Let d_{jt} be the number of controller messages agent j read at that update and y_{jt} an indicator that it ends on the target. We report three slopes of y on d :

raw no adjustment; pools across budgets, rounds and update order.

population adjusted d and y residualised on (episode $\times$ round), so the comparison is between agents facing the same board in the same population at the same moment.

order adjusted additionally residualised on `micro_slot_index`, removing the dilution trend.

Board sampling is uniform and excludes self-authored messages, so after these adjustments the remaining variation in d is close to which messages an agent happened to draw. Residual dose standard deviation is 1.20 against a raw 3.26, so the adjustments do not exhaust the variation. Intervals are a cluster bootstrap over initializations.

9.3 Result

Table 8: Change in the probability of adopting the controller’s target per additional controller message read, with 95% cluster bootstrap intervals over 60 initializations.

outcome	adjustment	slope	95% interval	n updates
same round	raw	+0.0206	[+0.0169, +0.0241]	130647
same round	population adjusted	+0.0054	[+0.0039, +0.0069]	130647
same round	order adjusted	-0.0002	[-0.0019, +0.0014]	130647
next round	raw	+0.0126	[+0.0092, +0.0159]	92413
next round	population adjusted	-0.0002	[-0.0013, +0.0011]	92413
next round	order adjusted	-0.0012	[-0.0025, -0.0001]	92413

The raw slope is 0.0206 [+0.0169, +0.0241]: reading one more controller message is associated with about two percentage points more adoption. Holding the population and round fixed cuts it to 0.0054 [+0.0039, +0.0069]. Removing the update-order trend as well leaves -0.0002 [-0.0019, +0.0014] — indistinguishable from zero, and tightly bounded rather than merely imprecise, on 130647 agent updates.

Repeating the test with the agent’s vote at the end of the *following* round as the outcome gives -0.0012 [-0.0025, -0.0001]. There is no delayed individual adoption either.

9.4 What this does and does not establish

It does not mean the controller has no effect. The population-level effects are large and causally established in Section 4. It means the effect **does not appear as a within-round, within-population dose gradient**: among agents sitting in the same population at the same moment, the ones who happened to read more of the controller were not more likely to adopt its target.

Two readings are consistent with that, and this archive cannot separate them.

Saturation. Exposure is near-universal — 99.4% of agents read at least one controller message when one is available, and the mean dose is 5.9. Everybody is already well past any threshold, so the margin carries no information even if the first message matters a great deal.

Mediation. If the controller works by shifting a few agents, who then post and shift others, the (episode $\times$ round) adjustment absorbs exactly that propagation channel by construction. Conditioning on a mediator removes the effect one is trying to measure. This reading fits the horizon evidence of Section 3.3: controlled branches track silence for roughly five rounds before diverging, which is the signature of a slow collective process rather than immediate per-agent persuasion, and a within-round contrast is blind to it.

The honest summary is that the simplest mechanical story — each controller message directly pulls its readers toward the target — is *not* supported at the margin, while the collective story remains open. Distinguishing them needs either an experiment that randomizes exposure within a population, or one that varies the vote attached to a fixed set of facts.

10 The account

Putting the pieces together, the following story is consistent with every table above.

1. The population is competent, and competence is the baseline to beat. Left alone these agents converge on the correct allocation: 95.0% truth in report-only, 68.7% in full communication.

Any controller is competing against a process that mostly works. This single fact reframes every efficiency number: the denominator of “how much did control help” is small by construction.

2. Intervening is mostly disruption, and disruption scales with volume. The shared component $M^{(0)}$ is negative in 8 of 9 comparisons and deepens monotonically with the posting budget. The causal IPW estimator confirms this is not selection: forcing an action lowers truth support at the next round and further by lag 5. The mixture term $D_{\text{KL}}(Q\|P_{\text{base}})$ accounts for most of the fitted trajectory cost, which is the same statement in divergence form — the bulk of the statistical displacement the controller produces is target-independent.

3. Yet the command does get through, weakly but measurably. $D^{(0)}$ is positive in 7 of 9 comparisons and reaches 15.3 points; command information reaches 0.0992 bits; the label-swap null is exceeded in most cells. So this is not the paired-archive picture of a purely target-blind perturbation. The controller transmits a real, small directional signal on top of a large non-directional one. The ratio of the two is what the efficiencies measure, and it is small.

4. The damage is not an information shortage. Evidence coverage rises by a factor of 1.7 to 5.3 under control and grows with budget, the false-target controller supplies about as much coverage as the truth-target one, and no agent ever holds a complete proof in any arm. So the controller is supplying genuine facts while making collective inference worse. What it adds besides facts is a salient repeated vote. The natural next step is to say that agents copy that vote, but Section 9 tests that directly and cannot find it: holding the population and the update order fixed, reading more controller messages does not raise adoption. So the damage is neither an information shortage nor, apparently, direct per-message persuasion. What remains is a collective route — the controller shifts a few agents, whose posts then shift others — which the horizon evidence supports and which the micro test is structurally unable to see. The one benign corner, full communication at $\rho = 0.75$ and $b = 3$, is where autonomous performance is weakest and the controller speaks least; raising the budget there reverses the sign.

5. “Following” is a trap. Following improves in every comparison, because it averages the two own-target gains and the false branch is very good at installing the false answer. A metric can rise while the thing one cares about falls. This is the clearest single argument in the report for keeping signed, outcome-specific effects rather than a scalar success score.

11 What would falsify this, and what is not established

- **The sample is 60 initializations.** Not 1382 trajectories, not $24\times$ that in votes. Every interval here is a 49–60 unit bootstrap and is correspondingly wide. The bootstrap intervals on fitted information and divergence scores are conditional on the fitted predictions: they resample held-out contributions, not the whole training pipeline, and so omit training variability.
- **The snapshot is incomplete and its own validation is marked invalid.** Four full-communication/ $\rho=1$ cells are missing, so the profile $\times$ persistence interaction is only identified at $\rho = 0.75$. Claim 4 above rests on a single setting and should be treated as a hypothesis to test, not a finding.

- **Divergence estimates are lower bounds from fitted critics.** A ratio of two lower bounds has no guaranteed ordering with respect to the true efficiency. Ratios are suppressed wherever the cost bootstrap touches zero.
- **The mechanism is not established, and the micro test is not decisive against it either.** Exposure is near-universal in this design, so a null at the margin is also what saturation would produce; and the (episode $\times$ round) adjustment absorbs the collective propagation channel by construction. Separating saturation from mediation needs a design that randomizes exposure within a population, or that varies the target vote attached to a fixed set of facts.
- **No thermodynamic reading is offered.** $\mathcal{C}_\pi$ is a statistical displacement in nats, not heat, and no reverse process has been justified.

Reproduction

```
.venv/bin/python -m rnd_init_metrics.run \
  --config configs/analysis/rnd_init_metrics/\
new_rnd_init_experiment.yaml --stage all
```

Source archive, opened read-only:

/Users/rsanchez/Projects/agents_control/new_rnd_init_experiment

Rounds table SHA-256: 24425dd979098fc480102dfd40ee2f2e